Problem Statement
Smart Home Device Management System

A smart home company wants to design a system to manage different types of smart devices installed in homes. All smart devices share some common properties such as:

- Device ID
- Device Name
- Power Status (ON/OFF)

However, different types of devices behave differently and consume different amounts of power. To implement this system, use **Inheritance and Abstract Classes**.

Step 1: Create an Abstract Class SmartDevice

Attributes:

- deviceId (int)
- deviceName (String)
- powerStatus (boolean)

Methods:

- void turnOn()
- void turnOff()
- abstract double calculatePowerConsumption()
- void displayDeviceDetails()

The **displayDeviceDetails()** method should display:

- Device ID
- Device Name
- Power Status
- Power Consumption (formatted to 2 decimal places)

Step 2: Create the Following Subclasses

a) SmartLight

Additional Attribute: wattage (double)

`Power Consumption = wattage × 5 hours`

b) SmartFan

Additional Attributes: speedLevel (int), baseWattage (double)

`Power Consumption = baseWattage × speedLevel × 3 hours`

c) SmartAC

Additional Attributes: temperatureSetting (int), powerRating (double)

`Power Consumption = powerRating × 8 hours`

The program must:

- Accept user input
- Create the appropriate device object
- Use runtime polymorphism
- Turn the device ON
- Display device details
- Print power consumption formatted to two decimal places

Input Format

- 1. Choice (Integer) — 1 → SmartLight, 2 → SmartFan, 3 → SmartAC
- 2. Device ID (int)

- 3. Device Name (String)

Depending on choice:

If choice = 1:

- wattage (double)

If choice = 2:

- speedLevel (int)
- baseWattage (double)

If choice = 3:

- temperatureSetting (int)
- powerRating (double)

Constraints

- 1 ≤ choice ≤ 3
- 1 ≤ deviceId ≤ 10^6
- 0 ≤ wattage, baseWattage, powerRating ≤ 5000
- 1 ≤ speedLevel ≤ 5
- 16 ≤ temperatureSetting ≤ 30
- Device name may contain spaces
- Power consumption must be displayed with 2 decimal places

Output Format

For valid input:

```
Device ID: <deviceId>
Device Name: <deviceName>
Power Status: ON
Power Consumption: <value>
```

For invalid input: No output

Sample Test Cases

Test Case	Sample Input	Sample Output
1	1 101 Living Room Light 20	Device ID: 101 Device Name: Living Room Light Power Status: ON Power Consumption: 100.00
2	2 202 Bedroom Fan 3 50	Device ID: 202 Device Name: Bedroom Fan Power Status: ON Power Consumption: 450.00
3	3 303 Hall AC 24 1500	Device ID: 303 Device Name: Hall AC Power Status: ON Power Consumption: 12000.00

Hidden Test Cases

Test Case	Input	Expected Output
1	1 111 Kitchen Light 15	Device ID: 111 Device Name: Kitchen Light Power Status: ON Power Consumption: 75.00

2	2 222 Guest Room Fan 4 40	Device ID: 222 Device Name: Guest Room Fan Power Status: ON Power Consumption: 480.00
3	2 600 Faulty Fan 6 50	(No output) Invalid Fan Speed > 5
4	1 555 Balcony Light 0	Device ID: 555 Device Name: Balcony Light Power Status: ON Power Consumption: 0.00
5	3 700 Cold AC 15 1500	(No output) Invalid AC Temperature (<16)